University Watch

The amazing work that researchers and scientists are up to at premier institutions across India

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Science and technology has always been important in India, and even today, the world is taking note of India – ISRO's monumental achievements with its space program and Indian scientists and innovators are at the forefront of global organisations. India has always held the torch of scientific and technological research high. However, it's not just our government and scientific organisations that are doing things to make us proud, but also our leading institutions. This article will look at just a few of the very interesting things happening in educational institutions across the country.

The value of research can not be underestimated

THE EYES OF A QUADCOPTER

Drones are the in thing now, and not just because they're cool. They're especially important when it comes to imaging an extensive building, or when an inspector wants to examine defects in, say, the iconic Bandra-Worli Sealink.

While panoramic mosaicing is in essence a solved problem, there are still problems – say, when there isn't too much difference between images, and not enough distinguishing features are viewed. A team of researchers from ViGIL Lab, Computer Science and Engineering Department, IIT Bombay has come up with a method to overcome these drawbacks and further extend the imaging capabilities of quadcopter cameras.

They have developed an algorithm to achieve their results with image panoramas. "First, the captured video contains a large number of images – these are summarised by our algorithm to get a small set of images that will encompass the complete scene under interest," explains Prof. Sharat Chandran, CSE Dept, IIT Bombay. He continues, "These images when placed at proper positions using the IMU (Inertial Measurement Unit) data present on the quadcopter will provide a final mosaic".

A big problem with current devices is power. Due to limitations of batteries and time required to cover scenes in detail, we are not able to image large scenes using just one quadcopter. In such cases, the team proposes the use of multiple quadcopters that can image a large scene collaboratively. According to Meghshyam Prasad, Research Scholar, IIT Bombay, "We divide the scene into multiple parts, and each part can be covered by one quadcopter. In this scenario every quadcopter should be able to reliably identify every other quadcopter. This can be achieved by placing fiducials on quadcopters. Fiducials are commonly placed in environments to mark a uniquely identifiable object in the scene. However, current fiducials, such as AR tags, are not designed to handle blur, which is intro-